An Open-Source, AI-Assisted Electronic Laboratory Notebook for the AI Transformation of Synthetic Biology Research

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Abstract

Self-driving laboratories (SDLs) are transforming modern research by integrating AI and robotics to accelerate experimentation. Within these systems, standardized documentation and interoperable data structures are essential to ensure reproducibility and traceability. However, many existing electronic laboratory notebooks (ELNs) remain commercial and insufficiently flexible for multidisciplinary synthetic biology and biofoundry workflows. Here, we present an open-source ELN platform built on Visual Studio Code (VS Code) and GitHub, enabling researchers to document experiments directly within an integrated development environment (IDE) while leveraging distributed version control for transparent collaboration. Laboratory records are written in Markdown using standardized templates grounded in biofoundry workflow and unit operation hierarchies, supported by extensions for automatic template generation, metadata updates, and structured sample ID management. We further developed the LabNote AI Assistant using retrieval-augmented generation (RAG) anchored to validated protocols, as well as an Open Notebook-based Biofoundry Copilot for collaborative knowledge retrieval. Beyond documentation, AI-assisted workflow analytics reconstruct experimental graphs and identify bottlenecks, underscoring the importance of a structured ELN framework. Ultimately, this platform lays the foundation for semantic and autonomous laboratories by integrating AI into an ELN environment that oversees the entire research process and efficiently analyzes research information expressed in natural language.

1 Introduction

Self-driving laboratories (SDLs), in which AI and robotics autonomously perform experiments, are rapidly accelerating scientific research [1]. In such environments, standardization of experimental protocols and data formats is essential to ensure reproducibility [2]. In synthetic biology, abstraction hierarchies of biofoundry workflows and unit operations have been proposed to support interoperable research practices, highlighting the importance of systematic recording systems for the research process [3]. Laboratory notebooks play a critical role in scientific practice: they preserve experimental memory, connect activities from planning to manuscript preparation, and serve as a coherent medium that resolves inconsistencies across research stages [4]. However, traditional paper laboratory notebooks (PLNs) face significant limitations in modern laboratories where large volumes of data are generated. PLNs become bulky, difficult to manage, and inefficient for indexing and searching. Their strictly sequential format is also unsuitable for documenting automated or parallelized experiments [5]. As research data increasingly shift toward digital formats requiring computational processing,

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conventional documentation methods lag behind, making the adoption of electronic laboratory notebooks (ELNs) inevitable [6].

ELNs promote good scientific practice by supporting protocol standardization, long-term data preservation [7], and ensuring data integrity and accessibility [8]. They enhance transparency and reproducibility through efficient management of extensive metadata [9], while reducing the burden of laborious and repetitive documentation tasks [10]. Nevertheless, most ELNs are commercial products with a relatively short average lifespan of approximately 7 years, often due to company acquisitions, market instability, or insufficient developer support [11]. Compared to open-source alternatives, proprietary ELNs are expensive and create strong vendor dependency [12]. Although ELNs have become indispensable productivity tools in industry, particularly in pharmaceuticals [13], their adoption in academia remains limited. Open-source ELNs [14–16] are especially critical in academic settings because of economic constraints, the need for customization, and the decentralized nature of research environments [17]. Moreover, many ELNs are tailored to specific disciplines [16, 17], while the growing demand for multidisciplinary research calls for more flexible systems capable of supporting diverse scientific domains [18, 19]. Open laboratory notebooks that enable collaboration across laboratories are also essential for accelerating global research progress [20]. However, conventional digital note-taking tools [21–24] remain inadequate for structured communication and systematic scientific discussion.

GitHub-based ELNs have emerged as a promising solution for enhancing scientific collaboration. Originally designed as a platform for code storage and sharing, GitHub is particularly valuable because it can function not only as an ELN repository but also as an effective communication medium. As an open platform, it does not require costly subscriptions, and researchers can selectively privatize repositories when needed. GitHub is accessible across a wide range of devices, and its tablet and mobile applications further enhance portability and ubiquity. In addition, GitHub supports collaboration throughout the entire research lifecycle, while its version control system enables transparent tracking of modifications and ensures reproducibility [25]. Despite these advantages, when GitHub has been adopted for laboratory notebook writing, its core repository features have often been underutilized, with the platform serving merely as a text editor [25]. In synthetic biology, a multidisciplinary field combining robotics, automation, and AI, ELNs capable of seamlessly connecting hardware and software workflows are essential [26]. Furthermore, recent advances in AI-integrated ELNs have dramatically improved research efficiency [27]. Therefore, developing ELNs that are both flexible and easily compatible with AI technologies has become increasingly crucial.

Building on GitHub's collaborative ecosystem, we differentiated our platform by authoring laboratory notebooks directly within Visual Studio Code (VS Code), an integrated development environment (IDE). This approach enables code used throughout the research process to be stored directly alongside experimental documentation. Using Git, a distributed version control system [28], notebook records are systematically stored, versioned, and shared through GitHub repositories. To standardize documentation in biofoundry-based synthetic biology research, we applied structured templates based on workflows and unit operations. In addition, we developed AI-assisted tools to improve the quality of research records and enhance user convenience. Through practical applications in pilot settings, we validated the utility of the proposed ELN platform for documenting DNA assembly workflows. Finally, we established an AI framework capable of extracting relevant information from experimental records, performing workflow performance analysis, and suggesting improvements to address workflow bottlenecks. These functions provide an important foundation for advancing laboratory autonomy.

2 Methods

The platform was designed around four principles: (i) storing laboratory records as plain-text Markdown within a version-controlled repository to ensure longevity and interoperability, (ii) aligning document structure with biofoundry workflow and unit operation hierarchies for synthetic biology, (iii) integrating AI assistance for drafting and knowledge retrieval while anchoring outputs to validated protocols, and (iv) enabling workflow-level analytics from structured ELN content. Extensions were implemented using the VS Code Extension API. `LabNote Lite` provides commands and chat-based actions to create experiment folders, insert workflow and unit operation templates, and update metadata (e.g., date and time). `LabSample` supports creation and management of sample, reagent, consumable, and equipment identifiers, with optional linkage to a LIMS database; referenced IDs are collected and displayed in a side panel scoped at folder or parent-directory level.

The `LabNote AI Assistant` backend was deployed as a serverless application on RunPod. A single container runs a FastAPI application together with Redis and Ollama. Requests are handled via RunPod's `/run` and `/status` APIs; the system is stateless, with all context supplied per request. A retrieval-augmented generation (RAG) pipeline indexes Standard Operating Procedure (SOP) documents from the `sop/` directory: Markdown content is embedded with *nomic-embed-text*, stored in Redis, and retrieved passages are injected verbatim into prompts and cited. Draft generation uses a multi-agent setup: Specialist agents produce candidate drafts with multiple large language models (default *llama3.1:8b*; optional *mixtral:8x7b*, *llama3.1:70b*, and a *gpt-oss:120b*), and a Supervisor agent evaluates and ranks outputs until quality thresholds are met. User feedback is captured for direct preference optimization (DPO): selected and edited outputs are logged via `/record_preference`; preference pairs are stored in a dedicated Git repository and used in an automated pipeline (TRL framework, gradient checkpointing, 8-bit optimizers) to produce GGUF models redeployed via Ollama. A judge model is used for head-to-head evaluation, with results visualized in dashboards. The `Open Notebook-based Biofoundry Copilot` ingests files, URLs, and text, normalizes content to Markdown, and builds a searchable knowledge base with hybrid retrieval (SurrealDB); access is controlled via OIDC and notebook-level sharing, and an MCP server exposes a read-only search gateway for external tools.

Workflow analytics were implemented as Claude "skills" that operate on the standardized ELN directory layout. `Sample ID Tracking` scans `labnote/` to extract unit operation-level input/output relations; `Workflow Diagram` converts these to Mermaid diagrams via an intermediate JSON schema and a Python script. `Consumables & Reagents Tracking` parses reagents, consumables, and method sections and matches items to local unit-cost records for JSON summaries. `Workflow Automation Assessment` scores each unit operation on a 0-3 hardware automation scale. `Lab Note Review & Reference Addition` checks template compliance and input/output consistency and supports addition of methodological references. Analyses were run in the same environment used for routine Claude-based tool use (skills executed against local labnote directories).

3 Results

3.1 Architecture of the ELN Platform Using VS Code and GitHub

VS Code is a lightweight, fast, open-source code editor that supports a wide range of programming languages and is available across macOS, Linux, and Windows operating systems. It provides an intelligent development environment through features such as syntax highlighting, bracket matching, auto-indentation, IntelliSense-based code completion, and integrated debugging tools. With built-in support for build and scripting systems, developers can perform most daily programming tasks directly within the editor. In addition, VS Code offers a rich extension ecosystem, enabling users to customize their development environment and contribute to the community by sharing their own extensions [29]. According to the 2025 Stack Overflow Developer Survey, VS Code is the most widely used development environment among developers worldwide [30].

Our ELN platform is designed such that research records are written in VS Code and uploaded to GitHub for sharing and collaboration (Fig. 1A). In this workflow, Git plays a central role as a robust version control system that stores every change made to source files and allows users to revert to previous states when necessary. Git is a distributed system in which complete version histories are stored locally, providing both reliability and flexibility. Because branches can be created and developed in parallel, multiple researchers can modify files simultaneously, facilitating collaboration and systematic conflict resolution. The Git workflow consists of four main stages: the working directory, the staging area, the local repository, and the central repository. The working directory is where files are created and edited in VS Code. Selected modifications are added to the staging area using the "git add" command, permanently recorded in the local repository with "git commit," and synchronized with the remote central repository on GitHub through "git push" [31]. VS Code further simplifies these processes by providing an intuitive graphical user interface (GUI) for Git operations.

The fundamental organizational unit of GitHub is the repository. Git manages version histories at the repository level and tracks all changes within the project directory. Repositories can be configured as either public or private, and access permissions can be controlled for individuals or

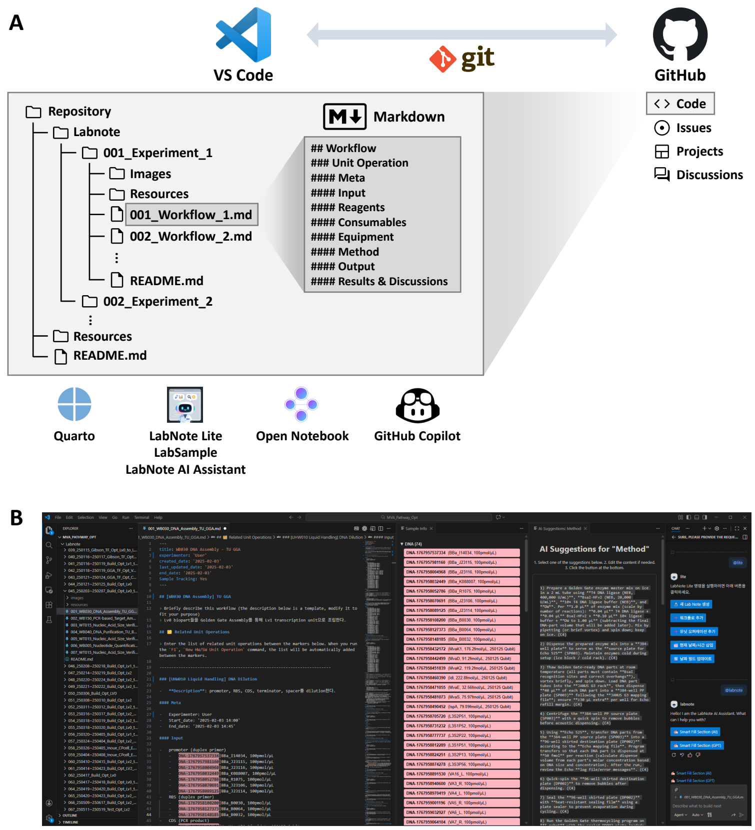


Figure 1: Architecture of the ELN platform based on VS Code and GitHub. (A) Overview of the ELN platform. (B) Example interface showing Markdown template-based experimental recording supported by dedicated VS Code extensions.

teams. Each repository contains the materials associated with a specific research project. Within the repository, the Code tab functions as both a source-code viewer and a file management system, hosting laboratory notebook files created in VS Code. Documents are organized hierarchically using a structured folder system: a "labnote" directory stores laboratory notebook entries, while a "resources" directory contains project-related materials. A README file provides essential repository information, including its purpose, methods, contributors, references, and directory structure.

Researchers collaborate through GitHub's integrated communication tools, including Issues, Projects, and Discussions. Issues are used to document planned experiments, unresolved problems, or project-related tasks. Features such as labels, milestones, and assignees support categorization and tracking. The Projects tab facilitates management of research progress and scheduling, allowing issues from multiple repositories to be consolidated within a single project dashboard. Researchers can discuss and update issues through comments, while workflow status can be assigned using stages such as "to do," "in progress," and "done," with optional start and end dates. Projects provide multiple organizational views, including tables, boards, and roadmaps, and issues can be filtered by criteria such as status, date, or assignee. Finally, the Discussions tab serves as a space for informal communication beyond structured research tasks, enabling the sharing of announcements, ideas, and broader scientific dialogue.

3.2 Standardized Document Structure for Synthetic Biology

Markdown is a lightweight markup language that converts plain-text documents into various formats, such as HTML. It enables the creation of structured documents using simple punctuation-based syntax. Because Markdown relies on intuitive symbols, it offers high readability and supports rapid writing, as formatting is applied directly during typing. As plain-text files, Markdown documents can be opened on any operating system or editor. Their simplicity also makes them highly compatible with version control systems, minimizing merge conflicts during collaborative editing. Markdown can seamlessly integrate diverse types of information, including images, mathematical formulas, code snippets, and tables, making it particularly suitable for laboratory notebook documentation. Markdown was adopted as the default format for GitHub README files, which has further contributed to its widespread popularity [32].

In our ELN platform, laboratory notebooks are organized into topic-based folders corresponding to individual experimental themes (Fig. 1B). Each experiment is recorded as a Markdown file using standardized templates grounded in workflow and unit operation structures designed for interoperable synthetic biology research within a biofoundry. The folder-level README file includes key metadata, experimental objectives, and a list of workflows associated with the experiment. Each workflow entry links directly to a corresponding Markdown file within the same directory. Within workflow files, researchers document unit operations and detailed experimental procedures in a systematic manner. The structure follows consistent Markdown conventions: workflow titles are written as second-level headings, unit operation titles as third-level headings, and the components of each unit operation as fourth-level headings. Unit operations include sections such as metadata, input, reagents, consumables, equipment, method, output, results, and discussion, with details recorded in bullet-point format.

In addition, image files and other experimental data are stored in dedicated "images" and "resources" folders, respectively. Images can be embedded directly within notebook files to document experimental observations. Related datasets and heterogeneous resources can also be conveniently linked, and the use of various VS Code extensions further enhances this process. This capability to connect diverse data types satisfies a key requirement for modern laboratory notebooks [33]. Although Markdown is simpler than many other markup languages, it may still present a learning curve for new users. To reduce this barrier, the Quarto extension's visual editing mode can be used to provide a more intuitive writing experience. This tool preserves the underlying text-based document format while offering visual editing features, such as buttons for styling, citations, tables, and image insertion. As a result, it provides a balanced approach between structured documentation and user-friendly editing [34].

To further support convenient and standardized notebook writing, we developed VS Code extensions aligned with the proposed template structure. The `LabNote Lite` extension simplifies the creation of experiment folders and workflow-based notebook entries, while ensuring consistent metadata recording. When semantic structuring is applied to an ELN, research data become interconnected through meaningful relationships, forming a scientific knowledge system that enhances research efficiency, reproducibility, and data reuse [35, 36]. Furthermore, integrating ELN-recorded workflows with a Laboratory Information Management System (LIMS), which manages samples, reagents, and equipment, has been shown to reduce the gap between documentation and sample tracking [37]. The `LabSample` extension supports ID creation and management, auto-completion, and highlighting of samples used in experiments. It is also linked to a LIMS database for direct item input. Referenced identifiers are automatically collected and displayed in the interface, enhancing usability and reducing errors in large-scale experimental documentation.

3.3 AI-Assisted Tools for Enhancing ELN Functionality

IDEs were originally created to consolidate tools that improve developer productivity. However, they are now evolving into intelligent development environments through the integration of AI technologies, which reduce cognitive load and shorten feedback cycles. As a result, numerous AI-based extensions have emerged in this context [38]. Modern platforms such as VS Code, JetBrains IDEs, and cloud-based notebooks now provide inline code assistants, conversational chat interfaces, and context-aware refactoring tools that operate across entire workspaces rather than only within open files. GitHub Copilot is a prominent AI-driven coding assistant that offers features such as autocomplete, multi-line code synthesis from comments, docstring generation, test scaffolding, and natural-language chat support. VS Code serves as the primary IDE environment for Copilot. By leveraging project-wide context including imports, symbols, and file structures, Copilot can propose edits or explanations, thereby accelerating routine coding and documentation tasks [39].

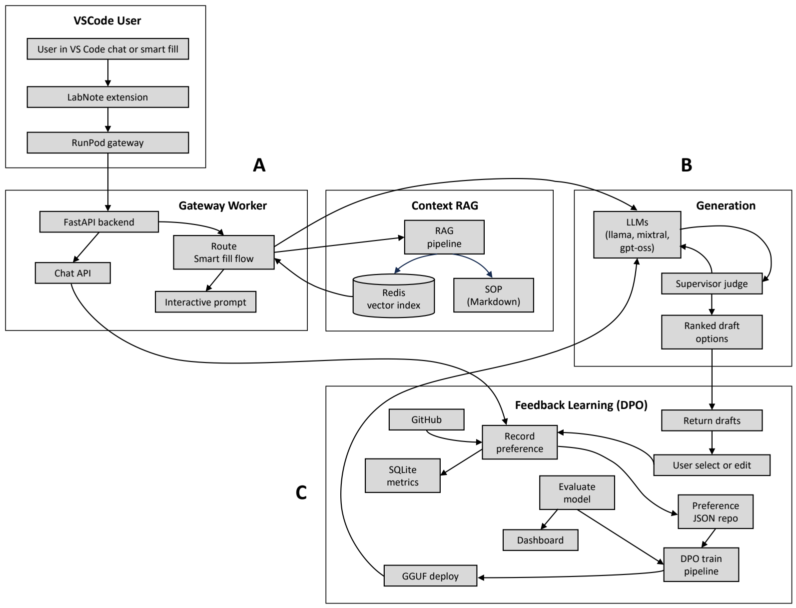
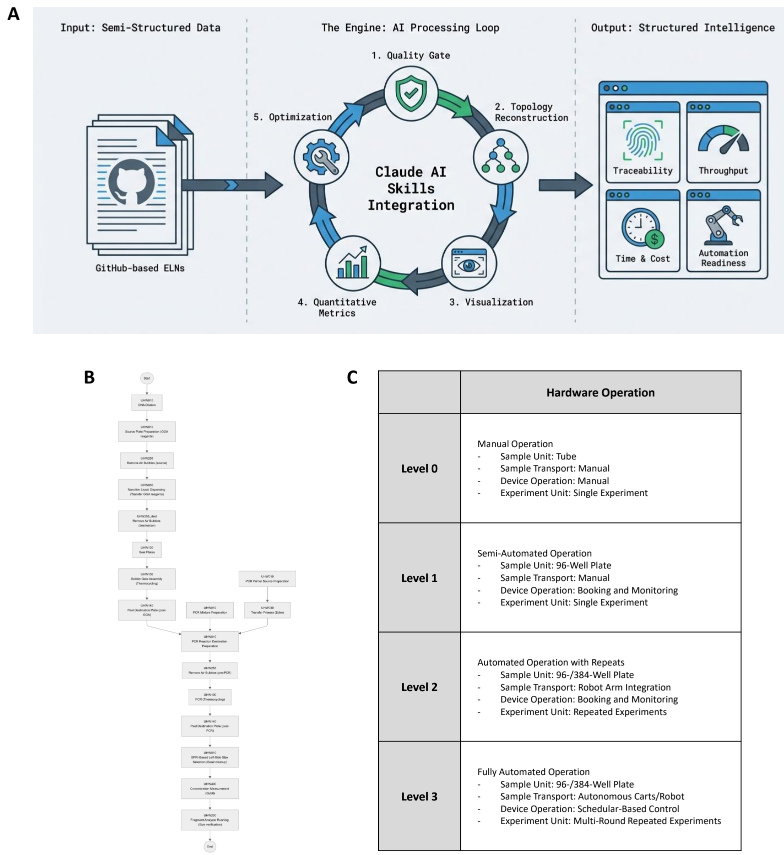


Figure 2: System workflow of the LabNote AI Assistant extension. (A) Serverless architecture connecting VS Code with a RunPod backend and RAG for protocol generation and section population. (B) Specialist-supervisor multi-agent loop for generating, evaluating, and ranking candidate drafts. (C) Preference logging via Git and DPO training pipeline with evaluation dashboards for continuous improvement.

Building on these advances, we developed the `LabNote AI Assistant`, an AI extension specifically designed to augment ELN workflows (Fig. 2). The tool minimizes manual effort during protocol drafting, metadata capture, and template enforcement, while ensuring that all AI-generated actions are logged and can be refined through user preference data. This design enables continuous system improvement while maintaining traceability and reproducibility of experimental records. The assistant enables researchers to draft protocol content directly inside unit operation templates. Users can invoke the tool through chat-based interactions, command palette actions, or slash commands (e.g., /populate <UO_ID> <Section>), allowing rapid completion of specific notebook sections. Generated outputs are guided by RAG, which references validated standard operating procedures to improve consistency with laboratory practices. Draft quality is further improved through iterative generation and evaluation, where multiple candidate drafts can be compared and refined. User selections and edits are captured as feedback signals, supporting systematic improvement of the assistant's drafting behavior across repeated use.

In addition, we developed an `Open Notebook-based Biofoundry Copilot` to support collaborative knowledge retrieval from heterogeneous sources in synthetic biology. The Copilot enables researchers to search not only across accumulated laboratory notebook records but also across related external materials, including uploaded files, URLs, and free-text resources. By normalizing diverse inputs into a unified Markdown-based representation, experimental knowledge can be systematically organized and reused within the ELN environment. The system employs hybrid retrieval strategies that combine keyword-based text search with semantic vector similarity search, allowing users to efficiently locate relevant protocols, contextual experimental information, and supporting references. During response generation, the Copilot can issue multiple retrieval queries and synthesize intermediate evidence into a coherent final answer, improving the quality of knowledge-assisted documentation and decision-making. To ensure secure collaboration, access to retrieved content is controlled through notebook-level sharing policies and authentication mechanisms, while external tools such as VS Code can interact with the system through restricted search interfaces.



	Hardware Operation
Level 0	Manual Operation - Sample Unit: Tube - Sample Transport: Manual - Device Operation: Manual - Experiment Unit: Single Experiment
Level 1	Semi-Automated Operation - Sample Unit: 96-Well Plate - Sample Transport: Manual - Device Operation: Booking and Monitoring - Experiment Unit: Single Experiment
Level 2	Automated Operation with Repeats - Sample Unit: 96-/384-Well Plate - Sample Transport: Robot Arm Integration - Device Operation: Booking and Monitoring - Experiment Unit: Repeated Experiments
Level 3	Fully Automated Operation - Sample Unit: 96-/384-Well Plate - Sample Transport: Autonomous Carts/Robot - Device Operation: Schedular-Based Control - Experiment Unit: Multi-Round Repeated Experiments

Figure 3: AI-assisted workflow performance analysis using structured ELN records. (A) Overview of the Claude skills-based workflow analytics pipeline. (B) Example of a DNA assembly workflow diagram represented in Mermaid syntax. (C) Assessment criteria used to determine automation levels of hardware operations.

3.4 Performance Analysis of the Synthetic Biology Workflow

To move beyond well-written laboratory notebooks toward actionable process optimization in biofoundry-enabled synthetic biology, we conducted a performance analysis of DNA assembly workflows recorded in our ELNs. The end-to-end performance of a workflow is not determined by a single protocol step, but rather by a chain of interconnected unit operations that collectively define throughput, time, cost, and automation feasibility. To identify bottlenecks and improvement opportunities based on objective metrics rather than expert intuition, we developed a set of AI-assisted analytical tools (implemented as Claude "skills") that directly leverage the standardized ELN structure described in earlier sections (Fig. 3A). These tools transform semi-structured experimental records into structured datasets that can be statistically summarized and compared across workflows. Specifically, the analysis focused on three dimensions: (i) traceability and workflow topology, (ii) throughput, time, and cost, (iii) automation readiness and bottleneck discovery.

A key requirement for workflow analytics is the ability to reconstruct experimental graphs that describe how inputs are transformed into outputs across sequential unit operations. Using standardized sample identifiers recorded in the ELN, workflows can be traced to determine when samples are generated, reused, or propagated into downstream products. Importantly, this approach captures not only linear pipelines but also branching and merging structures that frequently arise in combinatorial biofoundry workflows. Extracted relationships were visualized as interpretable workflow diagrams in Mermaid format (Fig. 3B), providing a clear representation of experimental progression.

Once workflow topology is established, throughput can be estimated at both workflow and unit operation levels. In DNA assembly workflows, throughput can be defined operationally as the number of assemblies designed, executed, and successfully verified through downstream persistence of sample IDs. In addition, workflow performance is strongly influenced by time constraints. Even with automated instrumentation, delays often arise from incubation periods, manual preparation steps, or dependencies between operations. Although exact timestamps may not always be available, structured metadata supports coarse-grained timeline estimation. Cost analysis was also supported through extraction of reagent and consumable usage from standardized notebook sections, producing itemized summaries that enable resource-based comparisons across workflows.

Beyond time and cost, workflow optimization depends on automation readiness, namely whether unit operations can be realistically scaled through hardware automation. Unit operations were evaluated on an automation scale ranging from manual to fully automated execution (Fig. 3C). The scoring rubric explicitly accounts for factors such as sample unit, sample transport, device operation, and experimental unit scalability. By combining workflow centrality with contributions to time, cost, and automation limitations, the analysis highlights bottleneck operations that disproportionately constrain overall throughput. It can generate improvement recommendations such as: (i) prioritizing automation of sample preparation steps, (ii) integrating design-to-execution mapping pipelines, or (iii) restructuring the workflow to reduce rework loops.

A persistent challenge in AI-assisted workflow analytics is reliability, as results are only as accurate as the underlying experimental records. Therefore, template compliance and consistency checks were applied to ensure reliable workflow reconstruction and to avoid misleading quantitative interpretations. This review component functions as a quality gate. Automated analytics must be preceded by template validation to prevent "false precision," where quantitative outputs appear rigorous but are derived from incomplete logging.

4 Limitations

This study has several limitations that should be considered when interpreting the results. First, the analysis is primarily based on qualitative, workflow-level observations rather than large-scale quantitative benchmarking or repeated controlled experiments. Therefore, the findings are intended to provide insights into usability and process characteristics, rather than statistically generalizable performance guarantees. Second, the effectiveness of the proposed AI-assisted tools depends strongly on the structure, completeness, and consistency of ELN records. Missing or inconsistently logged metadata may affect the accuracy of workflow reconstruction and metric estimation. In addition, time and cost estimates are derived from recorded or inferred metadata rather than direct instrumentation logs, which may introduce approximation errors. Finally, the framework has been validated mainly on DNA assembly workflows. Further adaptation may be required to extend its applicability to other experimental domains, larger datasets, or more complex laboratory pipelines. These remain important directions for future work.

5 Conclusions

The tools used in the ELN developed in this study are neither commercial nor proprietary, allowing broad and easy access for researchers. Because the system is intuitive, even beginners can adopt it without difficulty. The combination of VS Code, a well-established code editor, and GitHub, a widely used platform for code storage and collaboration, provides a practical advantage by enabling the integration of experimental documentation and computational code within a single environment. The Markdown-based structure, together with supporting extensions, allows diverse data formats including text, code, and images to be managed seamlessly within one integrated workspace. This capability supports the execution of the design–build–test–learn (DBTL) cycle in synthetic biology, where tight connections between wet-lab and dry-lab workflows are essential. Moreover, research activities can be systematically organized using GitHub's individual- and team-based collaboration features. Experimental outputs are stored and managed within repositories, while GitHub Projects enables efficient coordination among researchers. Git's version control ensures transparency, and its real-time accessibility across multiple devices further enhances collaborative research.

The hierarchical ELN structure aligned with biofoundry-based interoperable synthetic biology workflows contributes to the standardization of experimental practices by capturing research processes in a consistent and reusable format. Routine procedures become easier to organize and search, fulfilling the ELN's role as a long-term repository of experimental knowledge and ultimately improving reproducibility. Automatic template generation and metadata recording reduce repetitive documentation burdens. By minimizing typographical errors and guiding users through structured reporting, the system enhances overall record quality. In addition, AI-assisted drafting of laboratory notebook content improves time efficiency, reduces the workload of documentation, and supports decision-making throughout the research process. Because the tool can learn from user-selected and modified suggestions across multiple AI models, it has the potential to eventually propose laboratory-specific protocol optimizations.

Importantly, the ELN framework developed in this study is not limited to synthetic biology. Rather, it represents a versatile and extensible platform applicable across diverse research domains. Its accessibility and strong customizability lower the barriers to ELN adoption in academic environments. Additional AI tools can be readily integrated, allowing the system to evolve continuously through extension development and updates. As modern research increasingly moves toward multidisciplinary collaboration, this ELN provides a valuable foundation for supporting communication and knowledge sharing across scientific communities.

By integrating large language models (LLMs) into ELN environments, experimental workflows described in natural language can be systematically analyzed, enabling researchers to extract desired information and insights from large-scale records. Statistical workflow reports allow researchers to grasp workflow performance at a glance, build more efficient experimental pipelines, and facilitate cross-team communication. Transforming ELN records into analyzable structures also supports future extensions such as automated scheduling and queue optimization. Further work will be required to strengthen the integration of complementary databases such as LIMS and to develop software and hardware solutions that enable end-to-end automation from experimental design to execution and analysis. Nevertheless, this study provides an important foundation for building semantic and autonomous laboratories by integrating AI into ELN systems that digitize and manage the full research lifecycle. Because AI-driven learning is derived from structured ELN evidence rather than anecdotal experience, systematic recording of experimental processes using our platform becomes even more essential.

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A Appendix / supplemental material

Code and supporting data are included in the additional submission files.

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